

Does the Sharing Economy Technology Disrupt Incumbents? Exploring the Influences of Mobile Digital Freight Matching Platforms on Road Freight Logistics Firms

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In this paper, we study the disruptive power of the mobile digital sharing economy to the road freight logistics industry. New information technologies, such as the mobile internet, mobile payment methods, and GPS, have enabled platforms to match freight shippers' demand with carriers' supply by utilizing the convenience and mobility of smartphones. We empirically study the influence of the emergence of mobile digital freight matching platforms in the U.S. on the profitability and stock performance of two types of incumbent road freight logistics companies: freight arrangement companies (freight forwarders and brokers) and trucking companies (freight carriers). Since the mobile digital freight matching platforms mainly match small and mid-sized trucking companies with shipping demand, we expect the platforms to introduce interference (direct) competition to traditional freight arrangement companies and exploitation (indirect) competition to large trucking companies, which potentially raises operational challenges for both types of incumbents. We use Difference-in-Differences (DID) analyses to study the incumbents' profitability and the event study method to evaluate their stock performance. We find that after the advent of the mobile digital freight matching platforms in the U.S., the profitability of the traditional freight arrangement companies counter-intuitively remained unchanged. In contrast, the profitability of the large trucking companies increased. Furthermore, we examine the changes in sales and costs behind these profitability changes. Our dynamic analyses suggest a time lag of the influence of the mobile digital freight matching platforms. Besides, we find that the stock value of the traditional freight arrangement companies received a significant negative shock, whereas the stock value of the large trucking companies showed no significant reaction. This study contributes to understanding the disruptive power of an emerging technology innovation of sharing economy to firm performances in an increasingly competitive and innovative business environment.

Key words: competition; mobile digital freight matching; sharing economy; technology innovation

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1. Introduction

The thriving sharing economy has a disruptive power to alter daily lifestyles, consumer behaviors, and firm operations, even to reshape the industrial operating environment. With the great success of car-pooling and lodge-sharing via mobile applications, the mobile digital sharing economy has successfully expanded to the logistics industry in a form of digital freight matching platforms on the mobile terminal. For example, J.B. Hunt, a leading public trucking company, launched its mobile digital freight matching platform, J.B. Hunt 360, in November 2014 (Business Wire, 2014). In May 2017, an initially Texas-based on-demand mobile application, Uber Freight, was launched to provide matching services for both short-haul and long-haul trips. Since its launch, Uber

Freight has grown to over \$359 million gross bookings in 2018, with over 400,000 drivers, 36,000 carriers, and 1,000 shippers (Uber Technologies Inc, 2019).

An illustration of the service supply chain embedded in the road freight logistics industry is depicted in Figure 1. In the road freight logistics industry, freight brokers build the bridge between shippers, freight carriers, and consignees. A request for freight transportation may be initiated by the side of either shippers or recipients. To fulfill the transportation demand, freight brokers contract with freight carriers to arrange cargo pick-up and delivery. Information is shared between brokers and carriers, and the physical flow is moved from shippers to recipients through the transportation service provided by freight carriers. Freight brokers are the hub of communication and transactions during the entire logistics process, while

freight carriers are the ones that own trucks and supply cargo transportation to consignees for shippers.

Technology is disrupting how carriers and shippers connect (Lockridge 2019). With the emergence of the digital freight matching services, freight shippers' demand can match carriers' supply via either web- or mobile-based brokers (the platforms), improving both parties' efficiencies (Armstrong Associates Inc, 2016). Mobile digital freight matching platforms provide convenience and mobility for both shippers and carriers. For example, shippers can see transparent pricing and get real-time quotes, and carriers can simplify the load-searching process and get fast payments (Lockridge 2019). Mobile digital freight matching platforms rely on information technologies, such as the internet, smartphone, GPS, mapping, and mobile payment technologies, all of which have been well developed over the past several years (Armstrong Associates Inc, 2016).

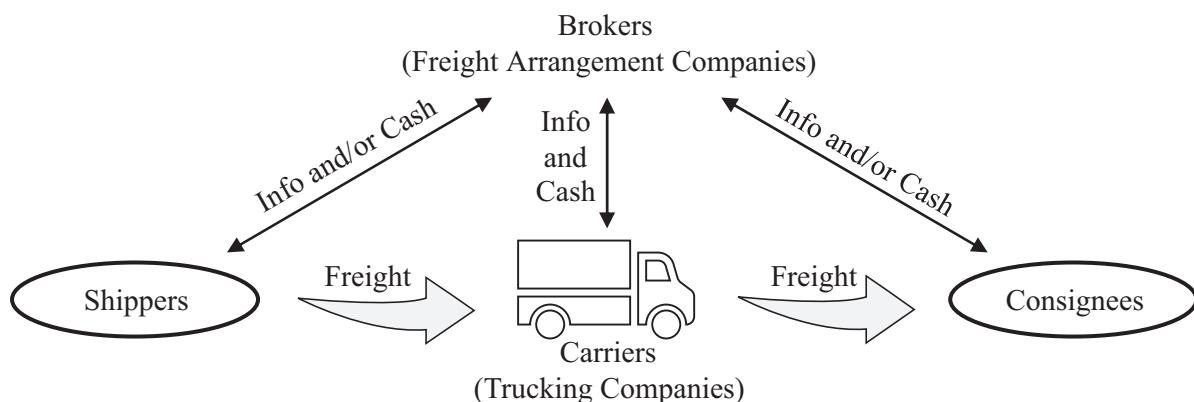
This study is theoretically motivated by the academic debates over the disruptive power of the sharing economy. On the one hand, many researchers have revealed the strong power of the sharing economy to reshape industries using ride-sharing as an example (Bai et al. 2019, Benjaafar et al. 2019, Burtch et al. 2018, Cramer and Krueger 2016, Guda and Subramanian 2019, Yu et al. 2019). On the other hand, researchers such as Christensen et al. (2015) posit that Uber is not a disruptive innovation since it did not originate in the low-end or new-market.

We believe whether an innovation is disruptive may also depend on the extent to which the companies in the industry have been affected by it. We suspect that mobile digital freight matching platforms introduced two types of competition to the logistics industry, potentially causing operational challenges to the incumbents. First, there is interference (direct) competition to traditional freight arrangement companies. During interference competition, companies

interact directly by fighting for common resources, such as customers. Freight arrangement companies are usually known as freight forwarders or brokers that arrange for cargo transportation by a freight carrier (NAICS Association 2018, U.S. Department of Transportation 2014). The emerging mobile digital freight matching platforms are expected to directly compete with traditional freight arrangement companies because the mobile platforms provide customers with substitutive services to incumbent traditional freight arrangement companies (Armstrong Associates Inc, 2016). Second, there is exploitation (indirect) competition to large trucking companies (freight carriers) that own the transportation assets and are in charge of the freight transportation (NAICS Association 2018, U.S. Department of Transportation 2014). During exploitation competition, companies interact indirectly by consuming common resources, such as truck drivers. The mobile digital freight matching platforms improve small freight carriers' performance by better matching their services with shippers. Although these small carriers may initially serve different customer segments than the large freight carriers, as the smaller companies grow with the help of the mobile platforms, they threaten large carriers by occupying potential resources, such as truck drivers. Hence, the mobile digital freight matching platforms create indirect competition to large trucking companies by helping small carriers. We are interested in understanding the extent of influence generated by mobile digital freight matching platforms on the traditional freight arrangement companies and large trucking companies.

This research is also practically motivated by the fast growth of mobile digital freight matching in the road freight logistics industry. The annual revenue of the digital freight matching brokerage market is expected to increase from \$11.2 billion in 2018 to \$52 billion by 2025 (Logisticsmgmt.com 2020). In addition,

Figure 1 Simplified Service Supply Chain Embedded in the Road Freight Logistics Industry



the trucking industry has long been contributing significantly to the U.S. national economy. In 2018, the trucking industry generated \$796.7 billion in revenue (American Trucking Associations 2019), representing 3.9% of U.S. GDP (U.S. Bureau of Economic Analysis 2019). Still, in 2018, 71.4% of the total national freight was transported by trucks, 3.5 million people were employed as truck drivers, and a total of 7.8 million employments were related to trucking activity that year (American Trucking Associations 2019). In this massive and growing industry, we are interested in the question: *what are the influences of mobile digital freight matching platforms on the profitability and stock performance of incumbent road freight logistics companies?*

In this study, we utilize the launch of the first mobile digital freight matching platform in the U.S. as a quasi-experiment to empirically study the intervention effect of the mobile digital freight matching platforms on the logistics incumbents. In particular, we consider the launch of MyDAT Trucker Services as the beginning of the mobile digital freight matching platforms in the U.S. and study its influence on the profitability and stock performance of road freight logistics companies. To the best of our knowledge, at the time of its launch in May 2011, MyDAT Trucker Services was the first mobile digital freight matching application in the United States. The application was designed to enable truck drivers to easily find nearby services, including available loads, truck stops, fuel locations, truck washes, and even Walmart stores (TransCore DAT, 2013). Since its initial nationwide launch in the U.S., the application has consistently been among the top 30 downloaded applications in the Google Play store's transportation category (TransCore DAT, 2013). Prior to the launch of MyDAT Trucker Services, any digital freight matching was simply web-based (Armstrong Associates Inc, 2016, Lockridge 2019). Dating from the 1990s, websites such as Truckstop.com, Getloaded.com, and directfreight.com, may have offered online services, but did not have the convenience and mobility that were later provided by smartphone applications.

Consistent with existing logistics studies (Fawcett et al. 2016), we study public road freight logistics companies to generate empirical insights for the logistics incumbents. Specifically, we conduct a Difference-in-Differences (DID) framework, which has been widely adopted in the research of logistics and operations management (Gallino and Moreno 2014, Song et al. 2018, Wan 2019), to investigate the influence of the mobile digital freight matching platforms on the profitability of road freight logistics companies – traditional freight arrangement companies and large trucking companies. Meanwhile, we apply the event study method on abnormal stock returns, because regressions in DID analyze mean

abnormal returns that may be largely influenced by outliers in the stock market (Jacobs and Singhal 2017). In the DID analyses on profitability, the observations are at the firm-quarter level between 2010 Q2 and 2016 Q1 (24 quarters). In the event study for stock market performance, we collect daily stock performance at firms between July 12, 2010, and June 8, 2011, including 210 trading days before the event day (the launch date of the first mobile digital freight matching platform in the U.S.), the event day, and 20 trading days after the event day. Quarterly financial data and daily stock prices are collected from the Compustat database. In total, our sample contains 888 firm-quarter profitability observations and 21,897 firm-day stock observations.

Our results show that, since the emergence of the mobile digital freight matching platforms in the U.S., the profitability of large trucking companies increased, but the profitability of traditional freight arrangement companies counter-intuitively remained unchanged. We explore the mechanism of these profit changes by analyzing the changes in sales and costs after the advent of mobile digital freight matching. After the emergence of the mobile digital freight matching platforms, both sales and costs of traditional freight arrangement companies decreased at a similar level, which resulted in no changes in profits. Meanwhile, both sales and costs of large trucking companies increased, and sales increased more than costs, leading to increased profitability. The rationale behind the changes could be that these traditional freight arrangement companies adopted strategic defensive actions such as cost-cutting to help to mitigate the interference competition, while large trucking companies employed aggressive growth-extension strategies to overcome the exploitation competition. Our dynamic analyses on the changes in profits also suggest a time lag of mobile freight matching platforms' influence. We also use Propensity Score Matching (PSM) to match samples in the treatment and control groups for robustness check. The estimated results using the matched sample by PSM show that all findings remain unchanged. Moreover, we find no significant reactions in the stock market for large U.S. trucking companies after the emergence of mobile digital freight matching platforms, but stocks of traditional freight arrangement companies in the U.S. received a significantly negative shock. Meanwhile, in the control group, stocks of both types of road freight logistics companies had no significant reactions, indicating the significantly negative shock only acted on the stock performance of traditional freight arrangement companies in the United States. Our findings help to understand the influence of the emerging mobile digital sharing economy and technology innovation on firm financial performance in

the increasingly competitive business environment. Our results not only suggest the disruptive power of the mobile digital sharing economy technology to the logistics industry, but reveal how an emerging technology innovation affects the performances of incumbents in a competitive and innovative business environment.

The remainder of the article is organized as follows: Section 2 gives a comprehensive literature review regarding our research topics. Section 3 reviews related theories and develops hypotheses. Section 4 introduces the mobile digital freight matching platforms and road freight logistics companies. Section 5 presents data, models, and results regarding profitability. Section 6 studies the effect on stock performance. Section 7 discusses our findings. Section 8 articulates the theoretical contributions and managerial implications. Section 9 presents conclusions, limitations, and future research directions.

2. Literature Review

Our research contributes to two streams of literature: technology innovation and sharing economy. This section first provides a general review of these two streams, and then emphasizes the technology innovation and sharing economy in logistics and supply chain management.

2.1. Technology Innovation

When production and operations management entered the era of Industry 4.0, innovations of digital technologies and automation technologies, such as cloud computing (Arbaban et al. 2021), AI and machine learning (Simester et al. 2020), blockchain (Babich and Hilary 2020), data analytics (Choi et al. 2018), and autonomous vehicles (Naumov et al. 2020), have reshaped many industries (Dai et al. 2020) and have caught scholars' attention.

Mobile digital freight matching applications mainly leverage the information advantages provided by the internet and mobile technologies. In recent years, more attention has been paid to information technology innovations as well (King et al. 1994, Lyytinen and Rose 2003, Swanson 1994, Yao et al. 2009a, Ye and Kankanhalli, 2018). Information technology innovation may also influence the competitiveness of companies (Bandyopadhyay et al. 2014, Li and Kumar 2018, Li et al. 2020, Niu et al. 2019, Wang et al. 2018). A summary Table of the literature on technology innovation is provided in Appendix S1.

Kumar et al. (2018) suggested that the operation management and information systems interface must be better understood in order to develop both disciplines. They also pointed out that the on-demand digital platform is one of the key areas that industries

and organizations must strategically grapple with. The insights provided by our study contribute to our understanding of the interface between operations management and information technology innovation.

2.2. Sharing Economy

Since the emergence of the sharing economy and innovative marketplaces in the past decade, many studies have contributed to this relatively nascent area (Hu 2020). Benjaafar and Hu (2020) reviewed the existing sharing economy literature and described three dominant applications: the P2P resource sharing platform, on-demand service platform, and on-demand rental networks. Based on user groups, a platform can also be classified into B2B, B2C, and C2C (Chen et al. 2020). Accordingly, the mobile digital freight matching platforms studied in this paper is a B2B on-demand service platform. Regarding the rest B2C and C2C platforms, Pei et al. (2021) have thoroughly explored how to react to internal and external sharing scenarios.

The literature of the sharing economy focuses on various topics, such as the pricing strategy (Cachon et al. 2017, Guda and Subramanian 2019, Jiang and Tian 2018, Taylor 2018, Tian and Jiang 2018), matching between customers and product or service providers (Bai et al. 2019), social welfare, consumer welfare, and environmental sustainability (Bellos et al. 2017, Benjaafar et al. 2019, Burtch et al. 2018, Yu et al. 2019), and even its conjunction with the blockchain (Cai et al. 2020).

The sharing economy has also been studied in many industries, including the well-explored passenger transportation industry (Bai et al. 2019, Benjaafar et al. 2019, Burtch et al. 2018, Chakravarty 2021, Cramer and Krueger 2016, Guda and Subramanian 2019, Steiner and Irnich 2020, Yu et al. 2019), tourism (Cui et al. 2020, Li and Netessine 2020), automobile (Bellos et al. 2017, Wan et al. 2019), and fashion product (Choi and He 2019). A summary Table of the literature on sharing economy is provided in Appendix S2.

We contribute to the stream on sharing economy by empirically studying the impacts of mobile digital freight matching platforms on the profitability and stock performance of various incumbent road freight logistics companies. We observe the performance of traditional freight arrangement companies and large trucking companies because the mobile freight matching platforms may generate different types of competition to them.

2.3. Technology Innovation and Sharing Economy in Logistics and Supply Chain Management

The technology innovations, such as drone operation (Agatz et al. 2018), blockchain (Choi 2019), boundary-spanning information technologies (Yao et al. 2009b), electronic logbooks (Miller et al. 2020, Scott et al.

2021), etc., introduce fundamental changes in logistics and supply chain management (Goldsby and Zinn 2016). As few studies have investigated sharing economy in the logistics and supply chain management context (Arslan et al. 2019, Qi et al. 2018), we contribute to technology innovation and sharing economy streams in the logistics industry. A detailed review in this domain and a summarized literature table are offered in Appendix S3.

3. Hypotheses Developments

In the road freight logistics industry, freight arrangement and trucking are two important segments. Freight arrangement companies are engaged in arranging freight transportation and providing services to carriers, shippers, and/or consignees, while trucking companies deliver cargo to consignees for shippers (NAICS Association 2018, U.S. Department of Transportation 2014). Consistent with existing studies (Fawcett et al. 2016), we focus on performance changes of public road freight logistics companies to generate empirical insights for the logistics incumbents. We expect that the mobile digital freight matching platforms generate different types of competition to these two types of companies in the freight logistics industry: (1) the interference (direct) competition to the incumbent traditional freight arrangement companies; and (2) the exploitation (indirect) competition to the incumbent large trucking companies. These two terms are borrowed from the ecosystem where the interference competition exists among species that interact directly by fighting for scarce resources, and the exploitation competition exists among species that interact indirectly by consuming scarce resources. The interference (direct) competition to the traditional freight arrangement companies and the exploitation (indirect) competition to the large trucking companies resulting from the emergence of mobile digital freight matching platforms are illustrated in Figure 2.

The mobile digital freight matching platform is an information technology innovation that combines the sharing economy concept with the technologies of mobile and wireless solutions. Mobile digital freight matching platforms provide easy connections, load transparency, real-time updates, and convenient transactions. Mobile digitalization of freight matching platforms emerged in the freight arrangement segment (Cosgrove 2019) and generated interference competition to traditional freight arrangement companies. That is, these mobile platforms share the customer segments with existing traditional freight arrangement companies by providing freight arrangement services connecting carriers and shippers. Those platforms could attract customers that used to be served

by the traditional freight arrangement companies. Therefore, the profits of traditional freight arrangement companies may be negatively influenced.

HYPOTHESIS 1. *The profitability of traditional U.S. freight arrangement companies decreases after the emergence of mobile digital freight matching platforms.*

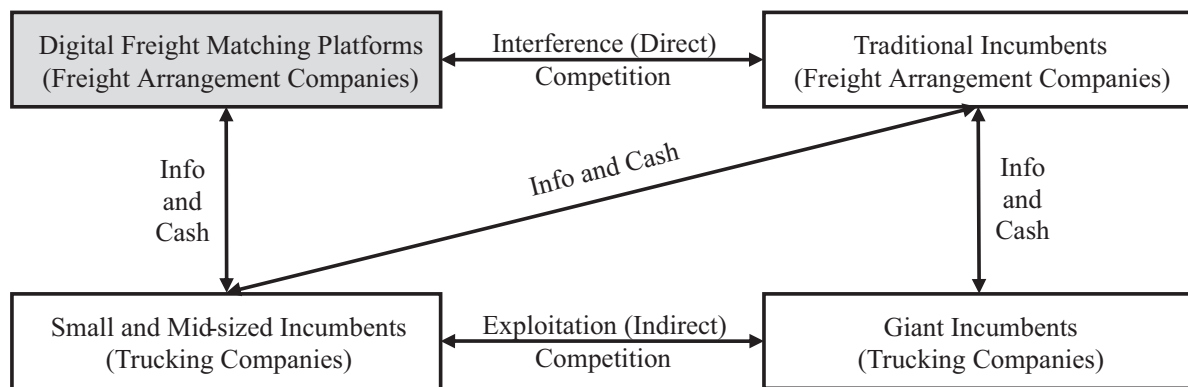
Mobile digital freight matching platforms focus on small and mid-sized carriers. (Dat.com 2020, Lockridge 2018). DAT, the launcher of the first mobile freight matching platform in the U.S., markets its product as the “small carrier’s secret weapon” (Dat.com 2020). In an interview with Uber Freight, an Uber Freight manager revealed that they primarily work with small carriers that own between one and fifteen trucks (Lockridge 2018).

These small trucking companies do not directly compete with large trucking companies because they serve different customer segments. Trucking companies are asset-based, and their operational capability is highly dependent on their tangible assets, which is a significant difference from freight arrangement companies. Large trucking companies enjoy well-known reputations and have more resources such as transportation equipment, terminal facilities, freight lanes, and transportation networks, while small carriers are limited by their low profile and smaller size (Macmillan 2019). Moreover, some shipping needs can only be fulfilled by large freight carriers.

By benefiting the small trucking companies, mobile digital platforms may generate exploitation competition with large trucking companies. If the small carriers keep growing with the help of the mobile digital freight matching platforms, they could compete with the large trucking companies by occupying common resources, such as truck drivers. Due to size differences, however, the competition for common resources is not immediately threatening. This enables large carriers to react to this exploitation actively. When the smaller carriers begin to encroach on common resources, large carriers could adopt lean management practices to improve productivity and profitability. With the help of strategic response, the large carriers may even improve their profitability when facing exploitation competition from the mobile freight matching platforms.

HYPOTHESIS 2. *The profitability of large U.S. trucking companies increases after the emergence of mobile digital freight matching platforms.*

Based on the efficient market hypothesis (EMH), the effect of an event on shareholder value is reflected in the stock price (Jacobs and Singhal

Figure 2 Sketch Map of Business Relationships embedded in the Road Freight Logistics Industry

2017). Peress (2010) pointed out that stock returns are affected by product market competition. Moreover, Li and Zhan (2019) found that competitive pressures from the product market may increase the risk of a stock crash. The emergence of mobile digital freight matching platforms introduces interference competition to traditional freight arrangement companies. Thus, the stock performance of traditional U.S. freight arrangement companies after the advent of mobile digital freight matching platforms is expected to be negatively influenced by the increased perceived competitive pressures.

HYPOTHESIS 3. *The stock performance of traditional U.S. freight arrangement companies is negatively influenced by the launch of the mobile digital freight matching platforms.*

Since the emergence of mobile digital freight matching platforms only creates indirect competition between incumbent small & mid-sized carriers and large trucking companies, the negative influence may not appear on the stock performance of large U.S. trucking companies. In addition, companies' stock market performances are complicatedly associated with their financial performances. The investors make stock investment decisions not only based on the past financial statements of the company, but also take into account their future expectations on firm financial performance. The large carriers may even improve profitability when facing the exploitation competition by conducting a strategic response. In turn, their stock market reactions match their foreseeable future financial performance. Hence, the stock performance of large trucking companies is expected to be positively influenced by the emergence of mobile digital freight matching platforms.

HYPOTHESIS 4. *The stock performance of large U.S. trucking companies is positively influenced by the launch of the mobile digital freight matching platforms.*

4. Mobile Digital Freight Matching Platforms and Road Freight Logistics Companies

Armstrong Associates Inc (2016) and Davis and Lucido (2017), respectively, summarized 27 and 19 mobile digital freight matching platforms targeting the trucking industry. Based on their studies as well as information organized from multi-sources, we list the mobile digital freight matching platforms in the U.S. road freight logistics industry by 2016 Q1 in Appendix S4.

In Appendix S4, 22 mobile digital freight matching platforms are listed by their launch year and month. Mobile digital freight matching services began in the U.S. in 2011. Although the idea of truck sharing emerged in the U.S. as early as the late 1990s in a website-based format (Armstrong Associates Inc, 2016, Lockridge 2019), represented by Truckstop.com, Directfreight.com, and Getloaded.com, the websites did not offer the convenience and mobility provided by smartphones. To the best of our knowledge, at the time of its launch in May 2011, MyDAT Trucker Services was the first mobile digital freight matching platform in the United States. This conclusion is supported by the results of an industry study by Armstrong Associates Inc, 2016, reporting that most mobile applications were launched during 2011–2015. Hence, our identification strategy is to use the launch of MyDAT Trucker Services as a quasi-experiment to study the influence of the introduction of mobile digital freight matching platforms.

We use NAICS and SIC classification systems together to identify freight arrangement companies

and trucking companies. The detailed process is provided in Appendix S5.

5. Effects on the Profitability

5.1. Financial Data

Six years of quarterly financial data from public freight arrangement companies and public trucking companies in the U.S. and other advanced economies are collected from the Compustat database for 2010 Q2–2016 Q1. The DAT official website mentions that MyDAT Trucker Services was launched in May 2011, which is the second quarter of 2011; thus, we use 2011 Q2 as the cut-off of the beginning of the treatment effect of mobile digital freight matching platforms in the United States. The four quarters before 2011 Q2 will be treated as the pre-intervention period, and the twenty quarters on or after 2011 Q2 represent the treatment period. We visualize the observation period and the timeline of the emergence of mobile digital freight matching platforms, as shown in Figure 3.

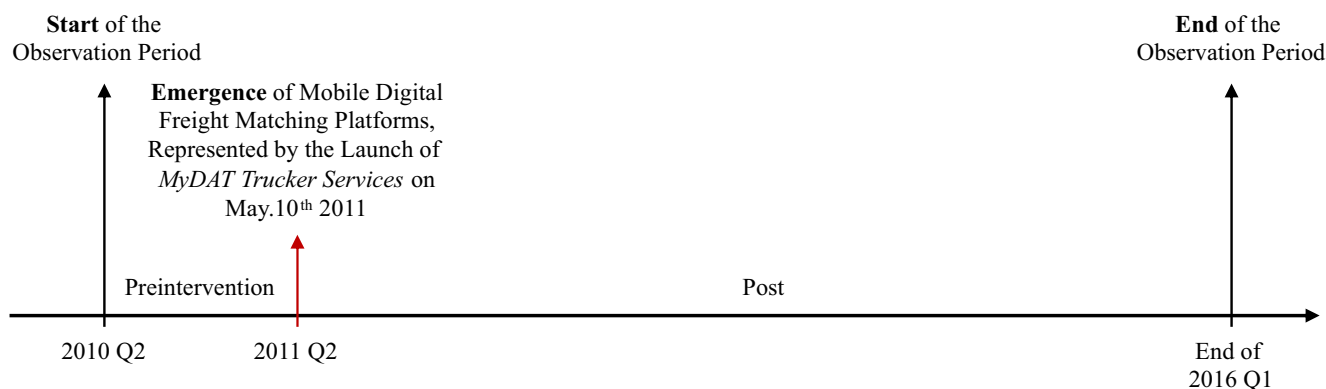
During our data window, digital freight matching on mobile terminals was mainly launched in the U.S. but not in many other countries. The U.S. companies are categorized in the treatment group, and we use companies from other advanced economies as the control group, because the U.S. macro-economic and political environments are more similar to the advanced economies than to developing economies. The included advanced economies are defined either by the United Nations, Human Development Index (HDI), International Monetary Fund (IMF), World Bank, or Organization for Economic Co-operation and Development (OECD). The included advanced economies should also have no mobile digital freight matching platforms launched before 2016 Q1. Since a mobile digital freight matching platform was launched in northern Europe as early as December

2011, represented by Courier Exchange (AppAdvice 2020), no economies from northern Europe are included in our samples.

From the Compustat database, we collect 22 U.S. public road freight logistics companies (including 7 freight arrangement companies and 15 trucking companies) in the treatment group, and 15 public road freight logistics companies from the advanced economies (including 12 freight arrangement companies and 3 trucking companies) in the control group. Overall, our samples include 37 public road freight logistics companies and 888 firm-quarter balanced observations. A list of the companies that are included in our data samples, along with their market countries and industry categorization information, is presented in Appendix S6. The selection process for these companies is explained in Appendix S5.

There are several tradeoffs between the length of the observation period, the included sample firms, and the global macro-economic environment. First, the observation period must be long enough to collect enough sample firms to test the dynamic effects of the intervention of mobile digital freight matching platforms. With a more extended observation period, however, fewer companies can generate balanced quarterly financial data. Second, the pre-intervention period should not include many macro-economic fluctuations, which may affect the identification of the impact of mobile digital freight matching platforms. There were several economic recessions in the 2000s both in the U.S. (i.e., the dot-com crisis in 2000–2002 and housing market crisis in 2006) and globally (i.e., the energy crisis in 2003 and the European debt crisis in 2009), which increase the fluctuations of firm performance and diminish the macro-economic matching between the treatment group and control group. These conditions warrant a longer post-intervention period, which in turn would reduce the pre-intervention

Figure 3 Observation Period and the Timeline of the Emergence of Mobile Digital Freight Matching Platforms [Color figure can be viewed at wileyonlinelibrary.com]



period. Considering the tradeoffs, we select 2010 Q2 – 2016 Q1 as our observation period.

The quarterly financial data that we collected from the Compustat database include *Operating Income Before Depreciation (Profits)*, *Sales*, *Total Operating Expenses (Costs)*, *Total Assets*, *Net Total Property Plant and Equipment (Net PPE)*, and *Total Liabilities*. *Profits* is used to measure firm-level profitability, while we also calculate *Return on Sales (ROS)*: the ratio of *Profits* to *Sales* and *Return on Total Assets (ROA)*: the ratio of *Profits* to *Total Assets* as replacement variables of *Profits* in the robustness check section. We use *Total Liabilities* and *Total Assets* to calculate *Asset-Liability Ratio (Leverage)*: the ratio of *Liabilities* to *Total Assets*.

Currency Exchange Rates for the Organization for Economic Co-operation and Development (OECD) member countries are collected from the OECD Statistics database; for non-OECD member countries, data are collected from ONADA, a New York-based international financial services company that provides internet-based forex trading and currency information services. *Quarterly GDP Change Percentage* for OECD member countries are collected from OECD Statistics; for non-OECD member countries, data are collected from tradingeconomics.com. Other yearly data such as *CPI Change Percentage*, *Gross Capital Formation (% of GDP)*, *Pump Price for Gasoline*, and *Population Density* are collected from the World Bank. The definitions of variables are summarized in Appendix S7.

Summary Statistics of quarterly financial data are provided in Table 1. Overall, the data structure of the treatment group is similar to that of the control group. We find that profits, sales, and costs in both treatment and control groups increase from the pre-intervention period to the treatment period. This indicates an increasing trend in the road freight logistics industry of both treatment and control groups. The change in profits in the treatment group is larger than in the control group.

Appendix S8 visualizes the mean plots of the profits for public road freight logistics companies from the U.S. and advanced economies during 2010 Q2 – 2016 Q1. The plots suggest that, before 2011 Q2, the profits of the companies in the treatment and the control group follow a parallel trend. However, the profit gap between the two groups increases after the intervention quarter. We also conduct statistical tests for the assumption of parallel trends in Section 5.3.

5.2. The Average Effects on Profits

We first develop the following DID model to investigate the effects of the intervention on the profitability of road freight logistics companies:

$$\begin{aligned} \text{Profits}_{it} = & c + \zeta_1 \text{Treated}_i \times \text{Post}_t \\ & + \zeta_2 \ln \text{Total Assets}_{it} + \zeta_3 \ln \text{Net PPE}_{it} \\ & + \zeta_4 \text{Leverage}_{it} + \zeta_5 \text{Currency Exchange Rate}_{it} \\ & + \zeta_6 \text{GDP Change \%}_{it} + \zeta_7 \text{CPI Change \%}_{it} \\ & + \zeta_8 \text{Gross Capital Formation (\% of GDP)}_{it} \\ & + \zeta_9 \text{Pump Price for Gasoline}_{it} \\ & + \zeta_{10} \text{Population Density}_{it} + \text{Firm}_i \\ & + \text{Year.Quarter}_t + \varepsilon_{it}, \end{aligned} \quad (1)$$

In Equation (1), Profits_{it} represents profits gained from regular operations before depreciation for company i in quarter t . The variable Profits_{it} has been adopted by operations management studies to directly measure profitability in OLS regression models (Kesavan et al. 2014, Khuntia et al. 2018). We do not apply the log transformation to the dependent variable as profits in some observations are negative. Treated_i is a dummy variable that equals 1 for companies in the treatment group, which captures the difference in profits between the treatment and control groups. Post_t is also a dummy variable that equals 1 if quarter t is on or after the quarter that the MyDAT Trucker Services mobile application was launched, and equals 0 if quarter t is before the launch quarter, which captures the difference in profitability before and after the start of the mobile digital freight matching platforms. $\text{Treated}_i \times \text{Post}_t$ is the DID term, capturing the difference in profitability before and after the emergence of the mobile digital freight matching platforms in the treatment and control groups. A positive coefficient of it (ζ_1) will suggest a positive treatment effect on profitability resulting from the emergence of the mobile digital freight matching platforms. Since the fixed entity effect for firm i and the fixed year-quarter effect for quarter t (Firm_i and Year.Quarter_t) are included in the equation, Treated_i and Post_t will not be identified and hence are omitted. Detailed definitions of all the above variables can be found in Appendix S7.

Some control variables are included. Total Assets_{it} is the value of total assets for company i in quarter t to control for firm size (Ba et al. 2013). Net PPE_{it} is the value of net fixed assets for company i in quarter t to control for logistics capital assets, such as buildings, land, furniture, and other physical capital that a business has purchased to run its business. Total Assets_{it} and Net PPE_{it} are log-transformed for a closer approximation of a normal distribution. Leverage_{it} , a variable used to control firm financial structure, is calculated by $\text{Total Liabilities}_{it}$ divided by Total Assets_{it} for company i in quarter t , which is also referred to as Asset-Liability Ratio. We have also controlled for macro-

Table 1 Summary Statistics of Quarterly Financial Performance: 2010 Q2–2016 Q1

Variable	Unit	Observations	Mean	SD	Minimum	Maximum
Panel A: Preintervention Period from 2010 Q2 – 2011 Q1						
– Treatment Group: U.S. Firms						
$Profits_{it}$	Million USD	88	27.36	35.61	–21.62	168.71
$Sales_{it}$	Million USD	88	312.63	368.88	7.94	1667.56
$Costs_{it}$	Million USD	88	285.26	341.92	8.42	1498.85
$Total\ Assets_{it}$	Million USD	88	665.79	723.85	10.14	2843.28
$Net\ PPE_{it}$	Million USD	88	316.53	379.28	0.11	1681.24
$Total\ Liabilities_{it}$	Million USD	88	357.24	591.47	4.02	2920.54
ROS_{it}	Rate	88	0.09	0.07	–0.08	0.29
ROA_{it}	Rate	88	0.04	0.02	–0.06	0.09
$Leverage_{it}$	Rate	88	0.48	0.21	0.22	1.11
$GDP\ Change\ \%_{it}$	Percent Point	88	0.48	0.44	–0.24	0.92
$CPI\ Change\ \%_{iy}$	Percent Point	88	2.02	0.66	1.64	3.16
$Gross\ Capital\ Formation\ (\% \text{ of } GDP)_{iy}$	Percent Point	88	18.83	0.16	18.74	19.10
$Pump\ Price\ for\ Gasoline_{iy}$	USD	88	0.76	0.00	0.76	0.76
$Population\ Density_{iy}$	People/sq. km	88	33.88	0.11	33.82	34.06
– Control Group: Advanced Economies' Firms						
$Profits_{it}$	Million USD	60	9.95	22.76	–7.11	111.27
$Sales_{it}$	Million USD	60	143.54	332.02	2.68	1475.28
$Costs_{it}$	Million USD	60	133.59	310.51	2.42	1364.02
$Total\ Assets_{it}$	Million USD	60	581.95	1319.32	17.18	5558.44
$Net\ PPE_{it}$	Million USD	60	177.20	418.21	5.06	1778.31
$Total\ Liabilities_{it}$	Million USD	60	250.11	516.43	5.93	2213.01
ROS_{it}	Rate	60	0.12	0.12	–0.24	0.46
ROA_{it}	Rate	60	0.02	0.01	–0.03	0.05
$Leverage_{it}$	Rate	60	0.48	0.14	0.29	0.77
$GDP\ Change\ \%_{it}$	Percent Point	60	4.10	8.77	–10.90	27.20
$CPI\ Change\ \%_{iy}$	Percent Point	60	3.42	1.66	1.52	8.57
$Gross\ Capital\ Formation\ (\% \text{ of } GDP)_{iy}$	Percent Point	60	22.62	4.08	13.54	31.27
$Pump\ Price\ for\ Gasoline_{iy}$	USD	60	1.48	0.57	0.23	2.52
$Population\ Density_{iy}$	People/sq. km	60	2554.40	3295.05	16.52	7363.19
$Currency\ Exchange\ Rate_{it}$	Rate	60	0.95	1.08	0.03	3.59
Panel B: Treatment Period from 2011 Q2 – 2016 Q1						
– Treatment Group: U.S. Firms						
$Profits_{it}$	Million USD	440	39.03	40.60	–3.67	203.37
$Sales_{it}$	Million USD	440	398.21	392.74	0.68	1768.83
$Costs_{it}$	Million USD	440	359.18	363.55	1.60	1599.21
$Total\ Assets_{it}$	Million USD	440	834.60	730.05	4.51	3098.56
$Net\ PPE_{it}$	Million USD	440	392.31	398.10	0.02	2098.16
$Total\ Liabilities_{it}$	Million USD	440	448.58	543.43	0.99	2947.27
ROS_{it}	Rate	440	0.10	0.11	–1.35	0.28
ROA_{it}	Rate	440	0.04	0.03	–0.12	0.10
$Leverage_{it}$	Rate	440	0.54	0.23	0.12	1.31
$GDP\ Change\ \%_{it}$	Percent Point	440	0.56	0.43	–0.28	1.35
$CPI\ Change\ \%_{iy}$	Percent Point	440	1.59	0.93	0.12	3.16
$Gross\ Capital\ Formation\ (\% \text{ of } GDP)_{iy}$	Percent Point	440	20.35	0.64	19.10	21.10
$Pump\ Price\ for\ Gasoline_{iy}$	USD	440	0.84	0.11	0.71	0.97
$Population\ Density_{iy}$	People/sq. km	440	34.61	0.37	34.06	35.30
– Control Group: Advanced Economies' Firms						
$Profits_{it}$	Million USD	300	12.60	18.96	–8.04	93.66
$Sales_{it}$	Million USD	300	255.08	541.58	1.63	3579.40
$Costs_{it}$	Million USD	300	242.47	528.16	1.83	3535.18
$Total\ Assets_{it}$	Million USD	300	783.67	1309.84	16.57	5229.86
$Net\ PPE_{it}$	Million USD	300	223.57	371.24	5.84	1701.75
$Total\ Liabilities_{it}$	Million USD	300	403.66	645.57	5.79	3071.27
ROS_{it}	Rate	300	0.13	0.12	–0.55	0.51
ROA_{it}	Rate	300	0.02	0.02	–0.10	0.06
$Leverage_{it}$	Rate	300	0.53	0.16	0.10	0.88
$GDP\ Change\ \%_{it}$	Percent Point	300	1.79	2.76	–4.40	10.90
$CPI\ Change\ \%_{iy}$	Percent Point	300	2.51	2.20	–0.63	8.89
$Gross\ Capital\ Formation\ (\% \text{ of } GDP)_{iy}$	Percent Point	300	23.16	4.64	12.83	31.27
$Pump\ Price\ for\ Gasoline_{iy}$	USD	300	1.57	0.58	0.22	2.54
$Population\ Density_{iy}$	People/sq. km	300	2674.09	3424.07	16.65	7908.72
$Currency\ Exchange\ Rate_{it}$	Rate	300	0.95	1.07	0.03	3.65

Note: 22 U.S. firms (7 freight arrangement companies and 15 trucking companies) and 15 advanced economies' firms (12 freight arrangement companies and 3 trucking companies) that have balanced data from 2010 Q2 – 2016 Q1 are included in the samples. U.S. firms of J.B. Hunt and C.H. Robinson are not included in the dataset in order to relieve endogenous concerns. Currencies of quarterly financial performances have been exchanged to USD based on the collected quarterly currency exchange rates.

Source: Quarterly financial data are collected from the Compustat database. Quarterly currency exchange rates for OECD member countries are collected from OECD Statistics; for non-OECD member countries, data are collected from ONADA. Quarterly GDP change percentage for OECD member countries are collected from OECD Statistics; for non-OECD member countries, data are collected from tradingeconomics.com. Yearly CPI change percentage, gross capital formation (% of GDP), pump price for gasoline, and population density are collected from the World Bank.

economic factors such as quarterly *Currency Exchange Rate* and *GDP Change Percentage*, and yearly *CPI Change Percentage*, *Gross Capital Formation (% of GDP)*, *Pump Price for Gasoline*, and *Population Density*.

We summarize the results in Columns (1) to (4) of Table 2. The DID term $Treated_i \times Post_t$ is first included in the model, then fixed effects and control variables are added step by step. In Column (4), we control for all variables as well as fixed firm effect and fixed year-quarter effect. In all of these estimations, we find that the coefficient of $Treated_i \times Post_t$ is consistently significant and positive ($p < 0.001$). The results suggest an increase in profitability for the U.S. road freight logistics companies at the significance level of 0.001 after the emergence of mobile digital freight matching platforms.

We further investigate the effects of the intervention on the profitability of traditional freight arrangement companies and large trucking companies, respectively. A Difference-in-Difference-in-

Differences (DDD) model is developed as follows:

$$\begin{aligned} Profits_{it} = & c + \theta_1 Treated_i \times Post_t \\ & + \theta_2 FreightArrangement_i \times Treated_i \times Post_t \\ & + \theta_3 \ln TotalAssets_{it} + \theta_4 \ln NetPPE_{it} \\ & + \theta_5 Leverage_{it} + \theta_6 CurrencyExchangeRate_{it} \\ & + \theta_7 GDPChange\%_{it} + \theta_8 CPIChange\%_{iy} \\ & + \theta_9 GrossCapitalFormation(\%ofGDP)_{iy} \\ & + \theta_{10} PumpPriceforGasoline_{iy} \\ & + \theta_{11} PopulationDensity_{iy} + Firm_i \\ & + Year_Quarter_t + \varepsilon_{it}, \end{aligned} \quad (2)$$

where $FreightArrangement_i$ is a dummy variable that equals 1 if the road freight logistics company i is a traditional freight arrangement company. The DDD term is $FreightArrangement_i \times Treated_i \times Post_t$,

Table 2 Impact on Profits for U.S. Road Freight Logistics Companies: Average Effects during 2010 Q2 – 2016 Q1

Variables	Coefficients				
	(1)	(2)	(3) <i>Profits_{it}</i>	(4)	(5)
$Treated_i \times Post_t$	23.883*** (2.242)	9.016*** (1.824)	9.352*** (1.826)	8.974*** (1.941)	12.288*** (2.086)
<i>Freight Arrangement_i × Treated_i × Post_t</i>					−10.212*** (2.491)
<i>ln Total Assets_{it}</i>			2.703 (1.889)	2.082 (1.910)	2.152 (1.892)
<i>ln Net PPE_{it}</i>			1.008 (1.408)	1.083 (1.414)	2.063 (1.421)
<i>Leverage_{it}</i>			−3.005 (3.993)	−1.046 (3.986)	−1.813 (3.952)
<i>Currency Exchange Rate_{it}</i>				58.477*** (11.752)	58.310*** (11.640)
<i>GDP Change %_{it}</i>				0.05 (0.142)	0.046 (0.140)
<i>CPI Change %_{iy}</i>				1.360+ (0.705)	1.290+ (0.699)
<i>Gross Capital Formation (% of GDP)_{iy}</i>				0.886** (0.299)	0.876** (0.296)
<i>Pump Price for Gasoline_{iy}</i>				−14.956* (6.345)	−15.640* (6.287)
<i>Population Density_{iy}</i>				0.016* (0.007)	0.015* (0.007)
<i>Constant</i>	15.148*** (1.578)				
Fixed Firm Effect	No	Yes	Yes	Yes	Yes
Fixed Year-Quarter Effect	No	Yes	Yes	Yes	Yes
Observations	888	888	888	888	888
Within R^2	0.114	0.247	0.253	0.278	0.292

Note: Standard errors are reported in parentheses. Samples include 37 public road freight logistics companies, with 22 from the U.S. (7 freight arrangement companies and 15 trucking companies) and 15 from the advanced economies (12 freight arrangement companies and 3 trucking companies) during 2010 Q2 – 2016 Q1. $Treated_i$ and $Post_t$ are included in column (1), but their coefficients are not displayed. $Treated_i$ and $Post_t$ in column (2) to (5) and $FreightArrangement_i$ in column (5) are omitted due to collinearity.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two-tailed tests).

representing the differences in the treatment effect of the mobile digital freight matching services between traditional freight arrangement companies and large trucking companies. $Treated_i$, $FreightArrangement_i$, and $Post_t$ are omitted due to collinearity. The estimated results are presented in Column (5) of Table 2. The coefficient of $FreightArrangement_i \times Treated_i \times Post_t$ is significantly negative ($p < 0.001$). This suggests that the traditional U.S. freight arrangement companies on average were less profitable than the large U.S. trucking companies at the significance level of 0.001 after the emergence of mobile digital freight matching platforms in the U.S.

This result demonstrates that profits increase more for large U.S. trucking companies than for traditional U.S. freight arrangement companies. This outcome may be explained by one of two possible scenarios: (1) profits for both types of companies increase, but the profits for large trucking companies increase more; (2) the profits for large trucking companies increase, but the profits for traditional freight arrangement companies do not change or even decrease. To explore which scenario actually occurred, we need to estimate the treatment effects on two types of companies, respectively. Accordingly, we split the dataset into two sub-datasets: one for traditional freight arrangement companies (456 observations) and the other for large trucking companies (432 observations), and estimate Equation (1) using two sub-datasets, respectively. The estimated results are reported in Table 3. The treatment effect of the mobile digital freight matching platforms is positive and significant at the 0.01 level for large trucking companies but not significant at the 0.1 level for traditional freight arrangement companies. These results support H_2 but do not support H_1 . We further explore the mechanism of these changes in Section 5.4.

5.3. Robustness Check for the Effects on Profitability

Various robustness checks have been adopted for the effects on profitability, including dependent variable replacement with ROS_{it} (Return on Sales) and ROA_{it} (Return on Assets), pre-intervention trend test, placebo tests, and Propensity Score Matching (PSM). All tests prove that our findings on profitability are robust. Details can be obtained in Appendix S10.

5.4. Unveiling the Mechanisms of the Changes in Profitability: Effects on Sales and Costs

In this section, we provide explanations for the results associated with profitability. The changes in profitability may be caused by different combinations of changes in sales (revenues) and costs. Three possible scenarios could lead to the unchanged profits of

Table 3 Impact on Profits for U.S. Traditional Freight Arrangement Companies and Large Trucking Companies: Average Effects during 2010 Q2 – 2016 Q1

Variables	Coefficients	
	Freight arrangement companies (1)	Trucking companies (2)
	<i>Profits_{it}</i>	
$Treated_i \times Post_t$	3.183 (2.088)	13.035** (4.375)
Firm-Level Control Variables	Included	Included
Macro-Economic Control Variables	Included	Included
Fixed Firm Effect	Yes	Yes
Fixed Year-Quarter Effect	Yes	Yes
Observations	456	432
Within R^2	0.212	0.409

Note: Standard errors are reported in parentheses. Column (1) includes 19 public freight arrangement companies, with 7 from the U.S. and 12 from the advanced economies during 2010 Q2 – 2016 Q1. Column (2) includes 18 public trucking companies, with 15 from the U.S. and 3 from the advanced economies during 2010 Q2 – 2016 Q1. $Treated_i$ and $Post_t$ are omitted due to collinearity.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$; **** $p < 0.001$ (two-tailed tests).

traditional U.S. freight arrangement companies: (1) increases in sales and costs at a similar level; (2) decreases in sales and costs at a similar level; or (3) no change in either sales or costs. Similarly, four possible scenarios may cause the increased profits of large U.S. trucking companies: (1) increases in sales and no change in costs; (2) increases in sales and decreases in costs; (3) increases in both sales and costs, with larger sales increases; or (4) decreases in both sales and costs, with larger cost decreases.

To explore which scenario actually explains the changes in profitability for traditional U.S. freight arrangement companies and large U.S. trucking companies, we replace the dependent variable in Equation (1) with $\ln Sales_{it}$ and $\ln Costs_{it}$, then estimate the equations for the treatment effects on sales and costs.

Like what we have done on profits, Appendix S9 visualizes the mean plots of the sales and costs for road freight logistics firms from the U.S. and advanced economies during 2010 Q2 – 2016 Q1. Again, we notice a parallel trend in the treatment group and the control group before 2011 Q2 for both sales and costs.

Table 4 shows the changes in sales and costs, respectively. Columns (1) and (2) analyze changes in sales and costs for the traditional U.S. freight arrangement companies. The results show that both their sales ($p < 0.05$) and costs ($p < 0.05$) are significantly decreased after the emergence of the mobile digital freight matching platforms. Specifically, after the launch of MyDAT Trucker Services in the U.S.,

Table 4 Impacts on Sales and Costs for U.S. Traditional Freight Arrangement Companies and Large Trucking Companies: Average Effects during 2010 Q2 – 2016 Q1

Variables	Coefficients			
	Freight arrangement companies		Trucking companies	
	(1) <i>ln Sales_{it}</i>	(2) <i>ln Costs_{it}</i>	(3) <i>ln Sales_{it}</i>	(4) <i>ln Costs_{it}</i>
<i>Treated_i × Post_t</i>	−0.133* (0.066)	−0.140* (0.066)	0.120** (0.039)	0.082* (0.042)
Firm-Level Control Variables	Included	Included	Included	Included
Macro-Economic Control Variables	Included	Included	Included	Included
Fixed Firm Effect	Yes	Yes	Yes	Yes
Fixed Year-Quarter Effect	Yes	Yes	Yes	Yes
Observations	456	456	432	432
Within <i>R</i> ²	0.674	0.665	0.753	0.711

Note: Standard errors are reported in parentheses. Column (1) and (2) include 19 public freight arrangement companies, with 7 from the U.S. and 12 from the advanced economies during 2010 Q2 – 2016 Q1. Column (3) and (4) include 18 public trucking companies, with 15 from the U.S. and 3 from the advanced economies during 2010 Q2 – 2016 Q1. *Treated_i* and *Post_t* are omitted due to collinearity.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$; **** $p < 0.001$ (two-tailed tests).

traditional U.S. freight arrangement companies' sales are decreased by 13.3%, and their costs are decreased by 14.0%. The decreases in sales and costs at a similar level explain the unchanged profits of traditional freight arrangement companies.

Columns (3) and (4) of Table 4 summarize the treatment effect on sales and costs for the large U.S. trucking companies. We find that after the launch of MyDAT Trucker Services in the U.S., large trucking companies' sales significantly increased by 12.0% ($p < 0.01$), and their costs also significantly increased by 8.2% ($p < 0.05$). Therefore, the increased profitability of the large U.S. trucking companies is explained by the fact that although both sales and costs increased, sales increase more than costs. Moreover, the decrease in the sales for the traditional U.S. freight arrangement companies supports the proposition that the emergence of the mobile digital freight matching platforms generated interference competition, while the increase in sales and costs of the large U.S. trucking companies provides support for the proposition that large trucking companies strategically reacted to exploitation competition caused by the emergence of these platforms.

5.5. Dynamic Effects

We also examine the dynamic changes of the effects of the mobile digital freight matching platforms on

the profitability of the U.S. road freight logistics companies and the sophisticated strategic decision-making process, consistent with the existing logistics and operations management literature (Song et al. 2018, Wan 2016). We notice a time lag of the influence of the mobile digital freight matching platforms, and the treatment effects become stronger when more mobile freight matching platforms are introduced to the market. Detailed analyses of the dynamic effects can be obtained in Appendix S11.

6. Effects on the Stock Performance

6.1. Stock Data

Data for the daily stock price of all the public road freight logistics companies in the U.S. and advanced economies that were not launched with mobile digital freight matching platforms are collected from the Compustat database. Data for the daily stock index of each included economy's stock market are collected from the investing.com website. Public road freight logistics companies from northern Europe and Canada are not included in the dataset. This is because the northern Europe market launched a mobile digital freight matching platform—Courier Exchange—just a half year later than the U.S. market, which is close to this first mobile digital freight matching platform launch event in the U.S. To avoid any potential endogenous factors that may have fluctuated their stock market, we exclude the northern European market. Canadian firms are also excluded because Canadian road freight logistics companies frequently transport or arrange freight shipping between the U.S. and Canada, so freight logistics events in the U.S. may potentially influence Canadian firms as well.

To precisely analyze the stock performance, we regard the release date of the MyDAT Trucker Services version 1.0 as the event day (Day 0), which is on May 10, 2011, based on AppBrain.com, a main Android app download platform. The observations contain 231 trading days between July 12, 2010 (Day -210), and June 8, 2011 (Day 20), including 210 trading days before the event day (the launch date of MyDAT Trucker Services in the U.S.), the event day, and 20 trading days after the event day. Following the existing studies on stock (Ba et al. 2013, Hendricks and Singhal 2009, Jacobs 2014, Jacobs and Singhal 2017), we measure stock performance by abnormal return, estimated by the market model (Brown and Warner 1985). This represents the proportion of a stock's return not explained by the change rate of the market index for the stock market. The market model employs the linear regression approach, using the daily stock market return as the regressor to estimate the daily stock return as follows:

$$R_{id} = \alpha_i + \beta_i R_{mid} + \varepsilon_{id}, \quad (3)$$

where R_{id} is the return of stock i on day d , calculated by the closing price on day d divided by the closing price on day $d - 1$ for stock i , and R_{mid} is the return of stock market index m where stock i is listed on day d , calculated by the closing price on day d divided by the closing price on day $d - 1$ for market index m . Since stock i is issued by firm i , our stock performance measurement is at the firm-day level. α_i is the intercept. β_i measures the relationship between R_{id} and R_{mid} . The proportion of the stock i 's return cannot be explained by the market movement is represented by ε_{id} .

Since our samples are collected from a wide range of advanced economies, we use the dominant index in each advanced economy to generate R_{mid} . For example, S&P 500 index, FTSE index, Hang Seng index, and KOSPI index are used for the U.S., Singapore, Hong Kong, and Korea's stock markets, respectively. Then, ordinary least squares (OLS) regression is employed to estimate α_i and β_i . Following the existing studies (Jacobs 2014, Jacobs and Singhal 2017), the estimation period starts from 210 trading days prior to the event day and ends at 11 trading days prior to the event day to separate the estimation and event periods to avoid any potential stock fluctuation caused by information disclosure (Hendricks and Singhal 2003). According to the literature, all included firms must have at least 40 stock returns during the 200-day estimation period in order to estimate the parameters correctly (Hendricks and Singhal 2009, Jacobs 2014, Jacobs and Singhal 2017). In addition, the included firms must also have at least one valid stock return after the event day. Therefore, 35 U.S. public road freight logistics companies are ultimately included in our stock performance samples, with 10 freight arrangement companies and 25 trucking companies; 68 advanced economies' public road freight logistics companies are included in the sample set, with 33 freight arrangement companies and 35 trucking companies. Our stock data samples, along with their industry categorization information, are presented in Appendix S12 (for U.S. firms) and Appendix S13 (for other advanced economies' firms).

Once we have the α_i and β_i estimated using daily stock and market index data with the market model Equation (3), adapted from the model Equation (3), abnormal return can be expressed by

$$AR_{id} = R_{id} - (\hat{\alpha}_i + \hat{\beta}_i R_{mid}), \quad (4)$$

where AR_{id} is the abnormal return of stock i on day d calculated from the market model. Detailed definitions of all the above stock variables can also be found in Appendix S7. Summary statistics of daily

stock performance are provided in Table 5. Our dataset includes 21,897 firm-day observations in total.

6.2. Statistical Testing on the Cumulative Abnormal Returns

We adopt the event study methodology to statistically test the cumulative abnormal returns. The DID analysis is a kind of parametric statistical approach that compares the mean value, so the results may be influenced by some extreme data values (Jacobs and Singhal 2017). Based on the summary statistics of daily stock performance in Table 5, we find that in the pre-intervention period, abnormal returns of the treatment group range from -0.48 to 0.85 with a mean value of zero, and those of the control group range from -0.47 to 0.27 with a mean value of zero. The DID studies using regressions may not be appropriate for such wide ranges in the abnormal returns of stock performance. However, event studies on stock performance normally also apply non-parametric statistics tests, such as the Wilcoxon signed-rank test to compare the median value, and the Binomial sign test to compare the percentage value of stock performance (Brown and Warner 1985, Hendricks and Singhal 2009, Jacobs 2014). Cumulative abnormal return (CAR) serves as the dependent variable used to measure the average effect of the event for the first mobile digital freight matching platform launch in the U.S., leveraging the event study approach to provide robust outputs. Since the stock market is sensitive to various events, one event can only show a limited length of intervention. In the existing event study studies, researchers tend to measure CAR in a relatively short period such as 2 days, 5 days, 10 days, or 20 days (Hendricks and Singhal 2003, Jacobs 2014), thus we will measure CAR up to around one trading month since the event day. Consistent with the existing event study research (Brown and Warner 1985, Hendricks and Singhal 2009, Jacobs 2014), we use the t -test to test for the significance of the mean abnormal returns that have adjusted for cross-sectional dependence, Wilcoxon signed-rank test to test for the significance of the median abnormal returns, and the binomial sign test to test whether the percent of negative abnormal returns are significantly different from 50%.

Since we are estimating the abnormal returns for multiple road freight logistics firms on the same event—the launch of MyDAT Trucker Services in the U.S.—the event dates of all sample firms are “clustered.” Clustered event dates can cause positive cross-sectional dependency (Brown and Warner 1985). To control for the clustering, the standard deviation should be estimated using the clustered version test (Brown and Warner 1985). Therefore, following

Table 5 Summary Statistics of Daily Stock Performance

Variable	Unit	Observations	Mean	SD	Minimum	Maximum
Panel A: 200-day Estimation Period: July 12, 2010 – April 25, 2011						
– U.S. Public Freight Arrangement Companies						
AR_{id}	Rate	2,000	0.00	0.04	−0.37	0.48
– U.S. Public Trucking Companies						
AR_{id}	Rate	4,884	0.00	0.05	−0.48	0.85
– Advanced Economies' Public Freight Arrangement Companies						
AR_{id}	Rate	6,302	0.00	0.03	−0.47	0.25
– Advanced Economies' Public Trucking Companies						
AR_{id}	Rate	6,560	0.00	0.02	−0.22	0.27
Panel B: Day 0 – Day 20 Post Period: May 10, 2011 – June 8, 2011						
– U.S. Public Freight Arrangement Companies						
AR_{id}	Rate	210	0.00	0.03	−0.17	0.24
– U.S. Public Trucking Companies						
AR_{id}	Rate	525	0.00	0.04	−0.25	0.28
– Advanced Economies' Public Freight Arrangement Companies						
AR_{id}	Rate	689	0.00	0.03	−0.15	0.12
– Advanced Economies' Public Trucking Companies						
AR_{id}	Rate	727	0.00	0.02	−0.10	0.09

Note: 35 U.S. firms (10 freight arrangement companies and 25 trucking companies) and 68 advanced economies' firms (33 freight arrangement companies and 35 trucking companies) that have a minimum of 40 stock returns during the 200-day estimation period as well as have stock returns after the event day are included in the samples.

Source: Abnormal returns are calculated from the market model based on daily stock price and daily stock market index. Daily stock prices are collected from the Compustat database. Daily stock market indexes are collected from investing.com.

the methodology built by both Brown and Warner (1985) and Hendricks and Singhal (2003), we use clustered version t -tests to determine the statistical significance of the mean abnormal returns. For a given time period (d_1, d_2) , the cumulative abnormal return (CAR) is:

$$CAR(d_1, d_2) = \sum_{d=d_1}^{d_2} \overline{AR}_d, \quad (5)$$

where

$$\overline{AR}_d = \sum_{i=1}^N \frac{\overline{AR}_{id}}{N}, \quad (6)$$

$\overline{AR}_d$ is the day d mean abnormal return of the N firms, and $CAR(d_1, d_2)$ is the cumulative abnormal return during day d_1 to day d_2 . For example, the cumulative abnormal return of the large U.S. trucking companies for the 21-day event period (0, 20) is the sum of each single day's average daily abnormal return of the included 32 firms. Moreover, the t -statistic TS_j for a j -day event period is:

$$TS_j = \frac{\sum_{d=d_1}^{d_2} \overline{AR}_d}{\hat{S}(\overline{AR}_d) \sqrt{j}}, \quad (7)$$

where

$$\hat{S}(\overline{AR}_d) = \sqrt{\frac{\sum_{d=-11}^{d=-210} (\overline{AR}_d - \bar{AR})^2}{199}}; \bar{AR} = \frac{\sum_{d=-210}^{d=-11} \overline{AR}_d}{200},$$

$\hat{S}(\overline{AR}_d)$ is the standard deviation of the mean abnormal return of the N firms for the 200-day estimation period, and $\bar{AR}$ is the mean abnormal return for the 200-day estimation period.

The results of the stock event study are presented in Table 6. Panel A-1 reveals the cumulative abnormal returns (CAR) for traditional U.S. freight arrangement companies. For traditional U.S. freight arrangement companies in the treatment group, the mean CAR is always insignificant at 0.05 level for various day combinations, although for the five-day period (16, 20), which is Day 16 through Day 20, mean CAR is significant at the 0.1 level. While, the median CAR for days (0, 5) is −3.09% and is significantly different from zero at the 0.05 level based on the results of Wilcoxon signed-rank test; and the median CAR for days (6, 20), days (16, 20), and days (0, 20) are −3.40%, −3.71%, and −7.74% and are significantly different from zero at the 0.05, 0.01, and 0.05 levels, respectively. Except for days (0, 2) in which only 40% of the sampled traditional freight arrangement companies (4 out of 10) have a negative CAR, at least 90% (9 out of 10) of the firms experience negative abnormal returns for days (0, 5), days (6, 20), days (16, 20), and days (0, 20), which are all significantly greater than 50% at the 0.05 level. Even better, for the five-day period (16, 20), 100% (10 out of 10) of the firms experience negative abnormal returns, significantly greater than 50% at the 0.01 level.

Panel A-2 presents the CAR for large U.S. trucking companies in the treatment group. No period has a

Table 6 Impact on Stock Performance for U.S. Traditional Freight Arrangement Companies and Large Trucking Companies: An Event Study Approach

Event days	Firm numbers	Mean ^a CAR	Median ^b CAR	% Negative ^c CAR
Panel A-1: U.S. Public Freight Arrangement Companies				
(0, 2)	10	-0.14%	0.57%	40.00%
(0, 5)	10	-4.26%	-3.09%*	90.00%*
(6, 20)	10	-5.16%	-3.40%*	90.00%*
(16, 20)	10	-6.19% ⁺	-3.71%**	100.00%**
(0, 20)	10	-9.49%	-7.74%*	90.00%*
Panel A-2: U.S. Public Trucking Companies				
(0, 2)	25	0.10%	0.03%	48.00%
(0, 5)	25	-1.58%	-0.72%	60.00%
(6, 20)	25	-1.72%	-0.10%	56.00%
(16, 20)	25	-0.83%	-0.17%	52.00%
(0, 20)	25	-3.30%	-0.74%	60.00%
Panel B-1: Advanced Economies' Public Freight Arrangement Companies				
(0, 2)	33	-0.93%	-0.41%	57.58%
(0, 5)	33	-2.30%*	-0.61%	60.61%
(6, 20)	33	0.55%	0.47%	48.48%
(16, 20)	33	1.02%	0.38%	45.45%
(0, 20)	33	-1.75%	-0.71%	60.61%
Panel B-2: Advanced Economies' Public Trucking Companies				
(0, 2)	35	-0.02%	-0.41%	57.14%
(0, 5)	35	-0.03%	-0.27%	57.14%
(6, 20)	35	0.04%	-0.88%	60.00%
(16, 20)	35	0.62%	0.50%	37.14%
(0, 20)	35	0.01%	-1.13%	57.14%

Note: Event day 0 is the launch date of MyDAT Trucker Services mobile application on May 10, 2011.

^a*t*-test is used to test the significance of the mean CAR.

^bWilcoxon signed-rank test is used to test the significance of the median CAR.

^cBinomial sign test is used when the sample size is greater than 25; Binomial distribution is used to determine the significance level when the sample size is less than or equals 25.

⁺ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two-tailed tests).

significant mean/median CAR or yield to negative CAR greater than 50% at the 0.1 level.

For comparison purposes, Panel B-1 and Panel B-2 are the results of stock event study for the traditional freight arrangement companies and large trucking companies from the control group. The only significant result is detected for mean CAR of traditional freight arrangement companies from advanced economies during days (0, 5). The high proportion of insignificant results in the control group indicates that the treatment effect is only limited to the U.S. road freight stock market, showing the robustness of the research design on stock performance as well as its outputs.

To summarize, except for the results of the very early three-day period (0, 2), non-parametric statistical tests on CAR of the traditional U.S. freight arrangement companies since the event day, with different length periods over the first trading month, provide evidence of significant and negative stock market reaction of traditional U.S. freight

arrangement companies to the launch of the U.S. first mobile digital freight matching platform. The insignificant results for days (0, 2) may be due to a short time lag of the reaction of the stock market to the launch of the first mobile digital freight matching platform. The above results demonstrate that the interference competition generated by the mobile freight matching platforms negatively affected the stock performance of traditional freight arrangement companies. However, the exploitation competition by the mobile freight matching platforms did not affect the stock performance of large trucking companies. Therefore, H_3 is supported, but H_4 is not supported.

6.3. Robustness Check for the Effects on the Stock Performance

We test the robustness of our method by obtaining the abnormal returns using the mean adjusted model instead, which defines that the abnormal return of stock i on day d is equal to the actual return of stock i on day d minus the average daily return of stock i from the 200-day estimation period. We also obtain abnormal returns by using 150-day and 250-day estimation periods versus our reported results that employ a 200-day estimation period, using the market model again. All findings remain consistent with those reported in Table 6. All these results suggest a significant and negative effect on the traditional U.S. freight arrangement companies since the first trading week after the event day, but an insignificant effect on the large U.S. trucking companies at any period.

We also use the DID approach to test the effects on the stock performance for both types of U.S. road freight logistics companies, but do not find significant results at the 0.1 level. Regressions in DID analyze the mean abnormal returns that may be largely influenced by outliers in the stock market (Jacobs and Singhal 2017). Detailed steps of DID analyses are provided in Appendix S14.

7. Discussions

Existing empirical studies have explored the application of sharing economy in various industries. To further extend the empirical research on the sharing economy, we use the launch of the first mobile digital freight matching platform in the U.S. to study the mobile digital sharing economy's disruptive power on various road freight logistics incumbents. We find that the introduction of mobile digital freight matching platforms disruptively affected the road freight industry. The impacts varied across companies' types (freight arrangement companies and trucking companies) and the performance measurements (profits and stock performance). Table 7 summarizes the hypothesis testing results and their associated mechanisms.

We first find that mobile digital freight matching platforms' emergence resulted in an increased average profit of the U.S. road freight logistics companies. Furthermore, this influence varied across two types of firms in the industry. Specifically, after the industry entered the era of mobile digital freight matching, the profits of the traditional U.S. freight arrangements remained unchanged, and the profits of the large U.S. trucking companies increased. Our results extend the existing research in other application contexts, where technology innovations can benefit focal firms and supply chain partners (Choi 2019, Niu et al. 2019), and where sharing economy platforms only benefit platforms, platform workers, product suppliers, and customers (Cachon et al. 2017, Choi and He 2019, Guda and Subramanian 2019, Tian and Jiang 2018). In our study, we notice that platforms may benefit certain business incumbents.

We further unveil the mechanisms of these changes in profitability. For traditional U.S. freight arrangement companies, both sales and costs decreased at a similar level, as shown in Table 4. However, for large

U.S. trucking companies, sales and expenses increased, but sales increased more than expenses. As for stock performance, traditional U.S. freight arrangement companies received a negative effect from the emergence of the mobile digital freight matching platforms, but large U.S. trucking companies and the road freight logistics companies in the control group did not. As existing sharing economy and crowdsourcing logistics studies talk about mechanisms mainly focusing on platforms' profit maximization problem (Bai et al. 2019, Benjaafar et al. 2019, Taylor 2018) and optimal resource efficiency (Arslan et al. 2019, Bellos et al. 2017, Qi et al. 2018), we also unveil the mechanisms how platforms can influence incumbents financial performance through different competition relationships.

The changes in profits, sales, and costs of the U.S. road freight logistics companies can be explained by the competition theory. The mobile digital freight matching platforms introduced competition to the road freight logistics industry. Meanwhile, when facing competition, companies may take strategic competitive actions to respond (Smith et al. 1989, Vannoy and Salam 2010). Lant et al. (1992) explained that managers tend to be motivated to react to the external competition if they believe the company's performance will be impacted. Hauser and Shugan (1983) argued that incumbents must defend their market position when they are facing new product and service threats. Since the mobile digital freight matching platforms brought interference (direct) competition to the traditional freight arrangement companies, the sales of those traditional freight arrangement companies would decrease due to more and more customers were attracted to the emerging platforms. In response to this competition as well as to the under-performed stock market, traditional freight arrangement companies may take defensive cost-cutting actions to ensure organizational survival under the threats of a potential recession. Hence, as the costs and sales were decreased at a similar level, traditional freight arrangement companies' profitability could remain unchanged. On the contrary, because the mobile platforms generated the exploitation (indirect) competition to the large trucking companies by helping small trucking companies, and due to the fact that stock performance did not decrease like freight arrangement companies, large incumbent carriers were motivated to adopt growth-extension strategies to prepare for potential competition in the future. Therefore, we observe that large trucking companies even increased profitability after the emergence of the new platforms. These results suggest that when facing competition, companies that take strategic actions may turn competition into a motivating force that helps turn the scale.

Table 7 Hypotheses Testing Results and the Associated Mechanisms

Hypotheses	Support	Mechanisms
<i>H₁ : The profitability of traditional U.S. freight arrangement companies decreases after the emergence of mobile digital freight matching platforms.</i>	No	I: Profits measure the difference between revenue (sales) and costs. As shown in Table 4, sales and costs of traditional freight arrangement companies decreased at a similar level, which led to unchanged profits.
<i>H₂ : The profitability of large U.S. trucking companies increases after the emergence of mobile digital freight matching platforms.</i>	Yes	II: As shown in Table 4, sales and costs of large trucking companies increased, and sales increased more than costs, which led to increases in profits.
<i>H₃ : The stock performance of traditional U.S. freight arrangement companies is negatively influenced by the launch of the mobile digital freight matching platforms.</i>	Yes	III: Direct competition was introduced to the freight arrangement segment by the launch of the first mobile digital freight matching platform, so rational stock investors held a bearish sentiment on the market value of traditional freight arrangement companies.
<i>H₄ : The stock performance of large U.S. trucking companies is positively influenced by the launch of the mobile digital freight matching platforms.</i>	No	IV: The impacts of the launch of the first mobile digital freight matching platform just indirectly influenced the trucking segment, and the negative effects on the freight arrangement segment did not spill over to the trucking segment either.

We also conduct analyses on changes in profits from a dynamic perspective and find a time lag for the impact to emerge. The lags may offer time for companies to invest effort into operational strategies. This result is consistent with a study done by Kalra et al. (1998), who found a significant time lag after the event for incumbents to react. Companies that react to the innovation of mobile digital freight matching platforms faster will gain more advantages over other companies in profit improvement.

In terms of the stock market, once the direct competition was introduced to the freight arrangement segment by the launch of the first mobile digital freight matching platform, rational stock investors may hold a bearish sentiment on the market value of traditional freight arrangement companies, and thus the stock performance of the traditional freight arrangement segment decreased. However, the negative effect on the stock performance of the freight arrangement segment did not spill over to the trucking segment. This fact is consistent with an event study done by Jacobs and Singhal (2017), who found no empirical evidence of any significant stock market reaction on downstream supply chain partners after a negative event happened to suppliers.

Our discoveries in profits and stock performance may be seemingly inconsistent: 1) the profitability of traditional freight arrangement companies was unchanged after the emergence of the mobile digital freight matching platforms, while their stock performance decreased; 2) profitability of large trucking companies increased, but their stock performance was not influenced. These results can be explained based on the theory of competitive dynamics (Lant et al. 1992, Smith et al. 1989, Vannoy and Salam 2010). The stock reactions to the launch of these platforms happened before the profitability change, so management teams could gain alerts from stock reactions and adopt different strategies. After seeing the decreased stock performance, traditional freight arrangement companies may adopt defensive actions by cutting costs, which reduced the supply and sales parallelly and led to unchanged profitability. The unchanged profitability can be an improved outcome compared to the potential profitability loss due to direct competition. Our empirical results in Table 4 support this inference by finding decreases in sales and costs of traditional freight arrangement companies. Meanwhile, the stable and unaffected stock performance for the trucking segment after the launch of these platforms indicated undetected threats to profitability, which may encourage the large trucking companies to employ growth-extension strategies in response to the exploitation competition, leading to increases in sales more than costs and ultimately an improvement in profits. These changes in sales and costs of large

trucking companies are also supported by our empirical results in Table 4. In short, the sensitive stock reactions and the chronological order that stock market reacts before business operations enable management teams to foreknow potential influences of mobile digital freight matching platforms by monitoring the stock market, so managers could adopt differentiated strategic competitive actions to improve financial performance, for example, reverse the possible negative situation (for traditional freight arrangement companies) and obtain the potential earnings (for large trucking companies).

8. Theoretical Contributions and Managerial Implications

8.1. Theoretical Contributions

This research contributes to several different literature domains. First, we contribute to the technology innovation stream of literature by exploring the disruptive power of the mobile digital freight matching platforms to various logistics companies through the lens of competition. Like the widely known industry 4.0 technologies, mobile digital freight matching is also proven to be a disruptive force in business operations. This disruptive energy can present at the operational level, financial level, firm-level, and industry segment-level. Our research shows that the emergence of one technology in one specific segment—for example, mobile digital freight matching in the freight arrangement segment—can profoundly disrupt other industry segments' performance.

Second, our study contributes to the literature on the sharing economy. In this study, mobile digital freight matching platforms have proven to be disruptive to various segments in the road freight logistics industry. Our research framework can be generalized to study sharing economy platforms in other industries. We also help to explore mechanisms behind the influence of sharing economy through competition. In addition, this study pays attention to the B2B multi-sided platforms that operate differently than the commonly studied C2C or B2C two-sided platforms.

Third, we contribute to the limited research on operations and financial performance in the logistics industry. The road freight logistics industry is vital to the U.S. macroeconomy and employment market. By studying the influence of mobile digital freight matching platforms on the profits and stock performance of road freight logistics companies, we shed light on improving financial performance under different forms of competition. The mechanisms explored in this research explain the different outcomes in profitability caused by technology innovations. By doing so, our study also responds to the call of Kumar et al.

(2018) to explore the interface of operations management and information systems.

Fourth, With the rapid development of disruptive technology innovation, industry-wide inter-firm competition is more diverse than ever. Competition theories offer theoretical foundations to explore nascent innovation phenomena. Previous scholars have linked topics of technology innovation with firm competitiveness (Bandyopadhyay et al. 2014, Li and Kumar 2018, Li et al. 2020, Niu et al. 2019, Wang et al. 2018). Our study contributes to competition theories by investigating the interference (direct) competition and exploitation (indirect) competition in the increasingly competitive and innovative business environment.

8.2. Managerial Implications

Our findings generate multiple managerial takeaways. First, the sharing economy technology has fundamentally changed operations in industries. While the applications of sharing economy have been well studied in the ride-sharing and lodge-sharing markets that are close to consumers, it remains unclear whether sharing economy renders a disruptive technology to B2B markets, such as the road freight logistics segment. Our findings offer insights into how sharing economy affects incumbents' performance in the transportation industry and provides suggestions to managers in these firms.

Second, this research alarms managers of technology innovation's disruptive power. The mobile digital freight matching technology is a specific innovation in the era of sharing economy and industry 4.0. With the coming industry 5.0, more technology innovations enhancing human and machine roles will be on the horizon. Findings in this study are not limited to the mobile digital freight matching technology or sharing economy but can be extended to future innovations.

Third, we highlight the influence of competition and the importance of strategic actions. In industry segments where technological innovation emerges and incumbents are directly competing with the new entrants, strategic defensive actions such as cost-cutting can help mitigate the competition. In our empirical setting, by reacting to the changing competitive environment strategically, traditional freight arrangement companies survived the interference competition without losing their profitability. Whereas, in segments that indirectly compete with the emerging technological innovation, aggressive growth-extension strategies can help incumbents perform even better. Large trucking companies took advantage of the aggressive strategic actions to overcome the exploitation competition and to improve their profitability. Thus, competition introduced by

the mobile digital freight matching platforms does not necessarily harm existing companies' performance in the road freight industry. Instead, the competition can be mitigated and alleviated, and may even help incumbents further improve their current business operations—if managers can dynamically and strategically react to the competition.

9. Conclusions, Limitations, and Future Research

Based on our study results, we can conclude that mobile digital freight matching—the mobile application of the sharing economy—generated significant impacts on companies in the road freight logistics industry. The emergence of this new technology negatively impacted the stock performance of the traditional freight arrangement incumbents in the short term but did not negatively impact their profitability. The mobile digital freight matching is even more beneficial to the giant incumbents in the trucking segment, as large trucking companies' profitability increased after the emergence of the mobile digital freight matching platforms, and their stock performance was not negatively influenced either. In sum, the mobile digital sharing economy in the road freight logistics industry has disruptive power for incumbents.

This empirical research has leveraged multi-methods and multi-data sources, but limitations remain and deserve future studies. First, there are other firms in the transportation and logistics industry in addition to the freight arrangement and trucking companies studied in this paper. Future research can further explore the influence on shippers and/or recipients that come from various manufacturing and supply chain industries. Second, we did not study the influences of mobile digital freight matching applications on the middle and small incumbents due to limited data sources. To extend the scope of the contribution, future research can adopt a survey approach to study the financial performance of U.S. middle and small road freight logistics companies. Third, the dimensions of performance we measured in this study are stock performance and profitability, as well as sales and costs, which have comprehensively represented the financial performances of road freight logistics incumbents. Operational measures can also be influenced by the emergence of mobile digital freight matching platforms. Future researchers with novel logistics operational datasets can dive into the effects of sharing economy or other disruptive technologies on firm operational performances.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix S1: Summary of Literature Review on Technology Innovation.

Appendix S2: Summary of Literature Review on Sharing Economy.

Appendix S3: Literature Review on Technology Innovation and Sharing Economy in Logistics and Supply Chain Management.

Appendix S4: Mobile Digital Freight Matching Platforms in the U.S. Road Freight Logistics Industry by 2016 Q1.

Appendix S5: Sample Selection Process.

Appendix S6: List of Road Freight Logistics Companies with Balanced Quarterly Financial Data: 2010 Q2 – 2016 Q1.

Appendix S7: Variable Definitions.

Appendix S8: Mean Plots of Profits for U.S. and Advanced Economies' Road Freight Logistics Companies: 2010 Q2 – 2016 Q1.

Appendix S9: Mean Plots of Sales and Costs for U.S. and Advanced Economies' Road Freight Logistics Companies: 2010 Q2 – 2016 Q1.

Appendix S10: Robustness Check for the Effects on Profitability.

Appendix S11: Dynamic Effects.

Appendix S12: List of U.S. Road Freight Logistics Companies Collected in the Event Study.

Appendix S13: List of Advanced Economies' Road Freight Logistics Companies Collected in the Event Study.

Appendix S14: Effects on the Stock Performance: Difference-in-Difference Approach.